CHAPTER 1:

(Part 1)

INTRODUCTION TO BUSINESS INTELLIGENCE

What is Business Intelligence?

Business Intelligence (BI) is a set of technologies and processes that allow people at all levels of an organization to access and analyze data.

Source: *Successful Business Intelligence* book.

What is Business Intelligence?

Business Intelligence is a set of methods, processes, architectures, applications, and technologies that gather and transform raw data into meaningful and useful information used to enable more effective strategic, tactical, and operational insights and decision-making.

Source: Forrester.com

What is Business Intelligence?

The term **Business Intelligence (BI)** refers to technologies, applications and practices for the collection, integration, analysis, and presentation of business information. The purpose of Business Intelligence is to support better business decision making. Essentially, Business Intelligence systems are **data-driven Decision Support Systems (DSS)**.

Source: Olap.com

What is Business Intelligence?

Business intelligence (BI) is a technology-driven process for analysing data and presenting actionable information which helps executives, managers and other corporate end users make informed business decisions.

BI encompasses a wide variety of tools, applications and methodologies that enable organizations to collect data from internal systems and external sources, prepare it for analysis, develop and run queries against that data and create reports, dashboards and data visualizations to make the analytical results available to corporate decision-makers, as well as operational workers.

Source: https://searchbusinessanalytics.techtarget.com/definition/business-intelligence-BI

Introduction BI - Summary



Evolution of Business Intelligence

- 1980s : Executive Information Systems (EIS), Decision Support Systems (DSS)
- 1990s : Data Warehousing (DW), Business Intelligence (BI)
- 2000s : Dashboards and Scorecards, Performance Management
- 2010 ++ : Analytics, Big Data, Personal BI, Data Science, Augmented BI, etc.

Source: The search for the perfect "business insight system", from Performance Dashboard, by Wayne Eckerson

http://download.101com.com/pub/tdwi/files/performancedashboards.pdf

Evolution of Business Intelligence

- Increasingly, however, business executives and workers are using business intelligence platforms themselves, as a result to the development of self-service BI and data discovery tools and dashboards. The BI market is expected to experience continuous growth as tools increasingly incorporates both Artificial Intelligence (AI) and machine learning (ML).
  - Self-service business intelligence (BI) is an approach to data analytics that enables business users to access and work with corporate data even if they do not have a background in statistical analysis, BI or data mining. Self-service BI tools allow users to filter, sort, analyze and visualize data without involving the organization's BI and IT teams.

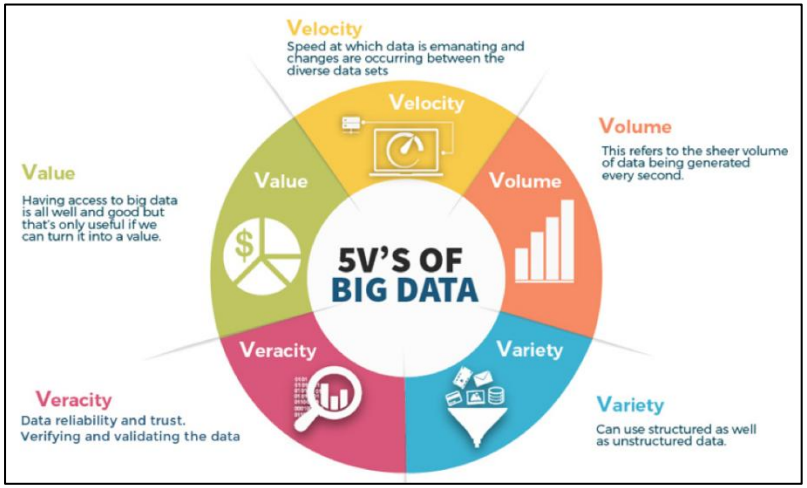
Concept, Effect of Business Intelligence in Business

- Overall, the role of business intelligence is to improve all parts of a company by improving access to the firm's data and then using that data to increase profitability.
- Companies that employ BI practices can translate their collected data into insights of their business processes.
- The insights can then be used to create strategic business decisions that improve productivity, increase revenue and accelerate growth.

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How does BI relates to Big Data?

- BI transforms data to assist in making informed decision.
- Big Data is a field that treats ways to analyze, systematically extract information from, or otherwise deal with data sets that are too large or complex to be dealt with by traditional data-processing application software.
- 5 V – Volume, Velocity, Variety, Veracity, Value



Source: https://www.techentice.com/the-data-veracity-big-data/

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How does BI relates to Big Data?

- Data sets – there are different types of data:
  - Numeric vs. textual
  - Structured vs. unstructured
  - Standard format vs. proprietary format
  - Internal vs. external data, system stored vs. file based data
  - Raw fact data vs. simulated/forecast/estimated data
  - Simple fact data vs. calculated metrics data

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Common Data Problems

- Structured, unstructured, semi-structured
  - Information and knowledge management is the management of both structured data (15% of information) and unstructured data (85% of information), according to the Butler Group.
  - 80% of business is conducted on unstructured information (Gartner Group).
- Information overloading
  - too much data and information with varied formats and structure
  - difficulty of data organization for effective access and retrieval
  - difficult to find useful information (knowledge) from them
  - Multiple copies of data exists sometimes with conflicts

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Common Data Problems – cont'd

- Data is everywhere
  - Data in separate systems and different sources; internal and external
  - Problem of spread-mart/spreadsheet data mart (http://en.wikipedia.org/wiki/Spreadmart)
  - Over 43 % of organizations have more than six content stores.

    (Forrester Research).
- Difficulty of access
  - We may have that data but we cannot access it (or difficult to get it), because of technical issues or administrative issues.
- Don't have that data
  - The data is simply not available.
  - The collection of data may need additional process and is costly.

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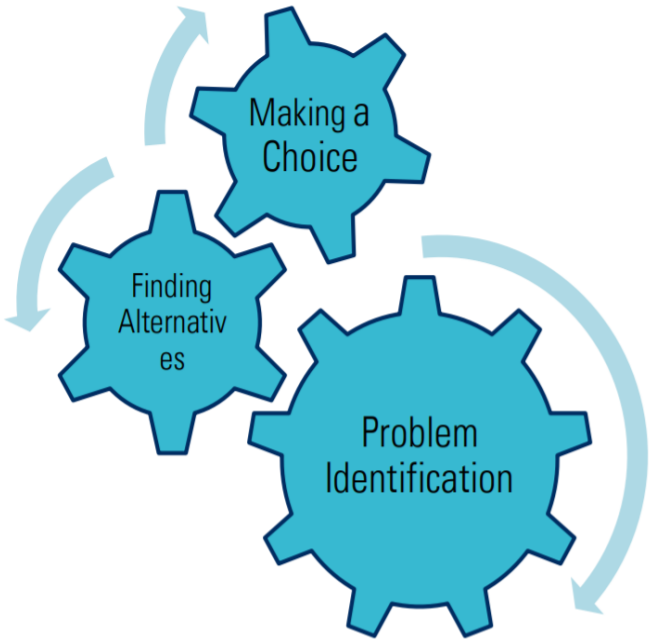
Decision Making Process

- The decision-making process, in a business context, is a set of steps taken by managers in an enterprise to determine the planned path for business initiatives and to set specific actions in motion.
- Ideally, business decisions are based on an analysis of objective facts, aided by the use of business intelligence and analytics tools.

Source: https://searchbusinessanalytics.techtarget.com/definition/decision-making-process

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Decision Making Process – cont'd



"Information and knowledge form the backbone of the decision making process"

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Decision Making Process – cont'd

- Decisions can be made based on:
  - Facts or data
  - Simulation (models)
  - Intuition, perception, sense
  - Group negotiation
- Traditionally, BI has been also understood as Decision Support System (DSS) – known as data driven DSS (data directly contributes to decision without intensive and advanced analytical techniques).

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Decision Making Process – Challenge

- A gap between data and knowledge (useful information leading to a decision).
- Management/operation by intuition.
- Lack of effective feedback and alignment systems, no improvement cycles.
- Need good analytical processing and models.

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Decision Making Process – Enhancement

- Evolving analytical is needed in decision support
  - ✓ Real-time, most recent data
  - ✓ Business user driven, agile, instant
  - ✓ Exploratory and interactive

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Decision Making Process – Steps

1. Identify a business problem.
2. Seek information about different possible decisions and their likely effect
3. Evaluate the alternatives and choose one of them
4. Implement the decision in business operations
5. Monitor the situation, gather data about the decision's impact and make changes if necessary.

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Decision Making Process – Who make a decision?

Decision making at different levels:

– Operational

- Related to daily activities with short-term effect
- Structured decisions taken by lower management

– Tactical

- Semi-structured decisions taken by middle management

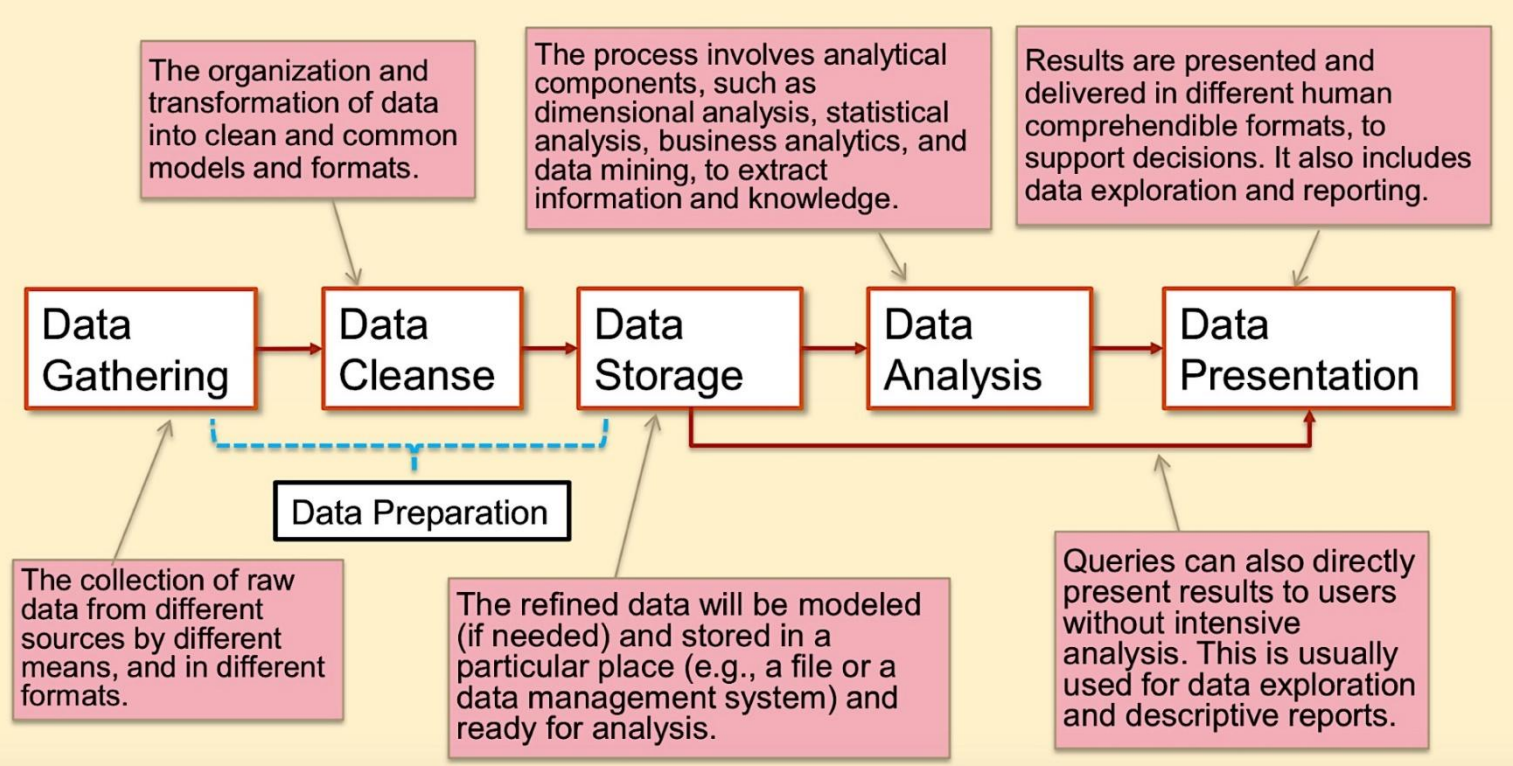
– Strategic

- Long-term effect
- Unstructured decisions taken by top management



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Business Intelligence – General Process



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BI General Process - Data Gathering and Cleansing

- Enterprise level data are coming from multiple different sources, but need to be combined and associated
  - Operational databases
  - Spreadsheets
  - Text, CSV
  - PDF, Paper

Data is never clean!
You will spend most of your time cleaning and preparing data!

- The need to bring together different data/information
  - Autonomous (may not have the control and management of data)
  - Distributed (from different systems and places)
  - Different (in data model, format, or platform)
- General processing steps - ETL
  - **Extraction**: accessing and extracting the data from the source systems, including database, flat files, spreadsheets, etc.
  - **Transformation**: data cleanse, change the extracted data to a format and structure that conform to the destination data.
  - **Loading**: load the data to the destination database, and check for data integrity
- Traditional BI focuses on upfront separate ETL processes that load the data in a centralized storage. In modern BI and analytics, data cleanse and transformation may happen just-in-time with analysis.

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BI General Process - Data Storage

- In traditional BI, a special database system called data warehouse or data mart is often used to store enterprise data
  - The purpose of a data warehouse is to organize lots of stable data for ease of analysis and retrieval.
- Traditional (operational) relational databases facilitate data management and transaction processing. They have two limitations for data analysis and decision support
  - Performance
    - They are transaction oriented (data insert, update, move, etc.)
    - Not optimized for complex data analysis
    - Usually do not hold historical data
  - Heterogeneity
    - Individual databases usually manage data in very different ways, even in the same organization (not to mention external data sources which may be dramatically different).
- The data warehouse approach is a centralized and structured approach for analytical data management. For more recent personal BI/analytics, data is also kept locally for easy access and manipulation, without much technical support.

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BI General Process - Data Analysis

- Descriptive reporting
  - Structured and fixed format reports
  - Based on simple and direct queries
  - Usually involves simple descriptive analysis and transformation of data, such as calculating, sorting, filtering, grouping, and formatting
  - Ad hoc query and reporting
- OLAP (Online Analytical Processing)
  - Focus on descriptive reporting
  - An optimized multi-dimensional analysis technology
- Business analytics
  - Business analytics (BA) is the practice of iterative, methodical exploration of an organization's data with emphasis on statistical analysis.
- Advanced and computation intensive: data mining, deep learning, etc.
  - Data mining techniques are a blend of statistics and mathematics, and artificial intelligence and machine-learning.

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BI General Process - Data Analytics

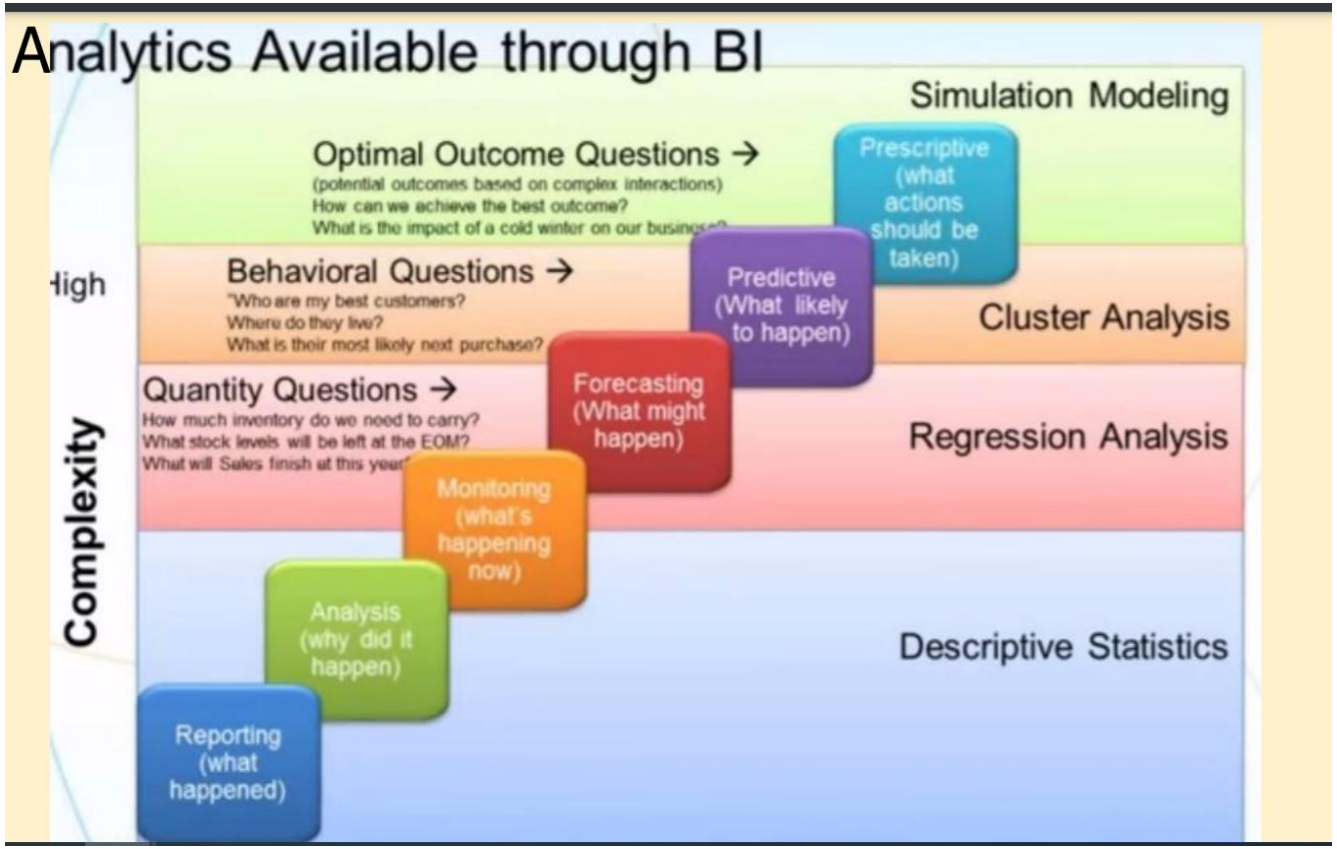
- Analytics refers to a more systematic, automated, and flexible process of data analysis for revealing insights and decision support in more extensive application areas, e.g. sports, disease, network traffic, etc.

  (http://pestleanalysis.com/differences- between-business-analytics-and-business- analysis/)

- Analytics initially referred to advanced statistical modelling using tools like SAS and SPSS. Now, analytics refers to the entire domain of leveraging information to make smarter decisions. In other words, reporting and analysis.

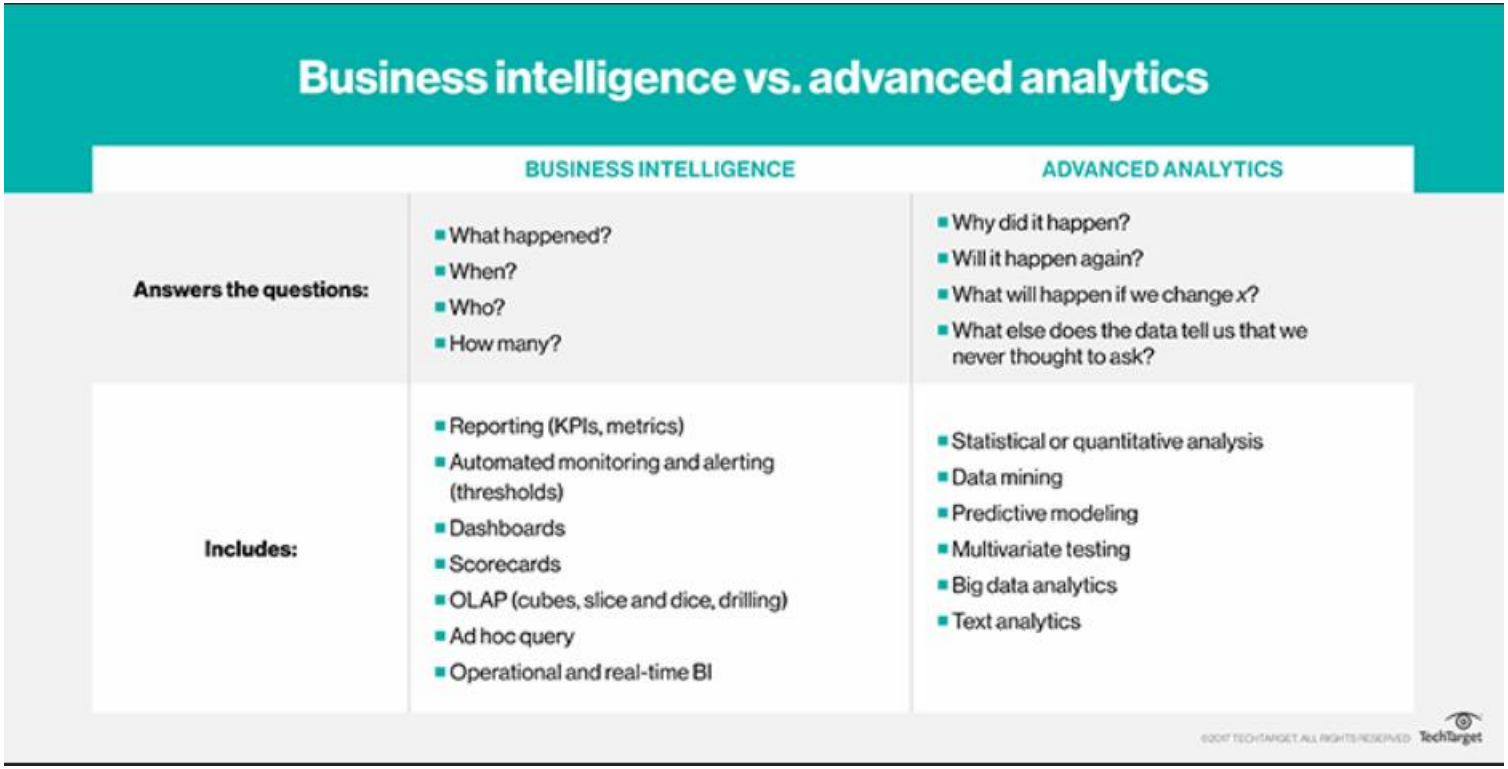
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BI General Process - Data Analytics (cont'd)



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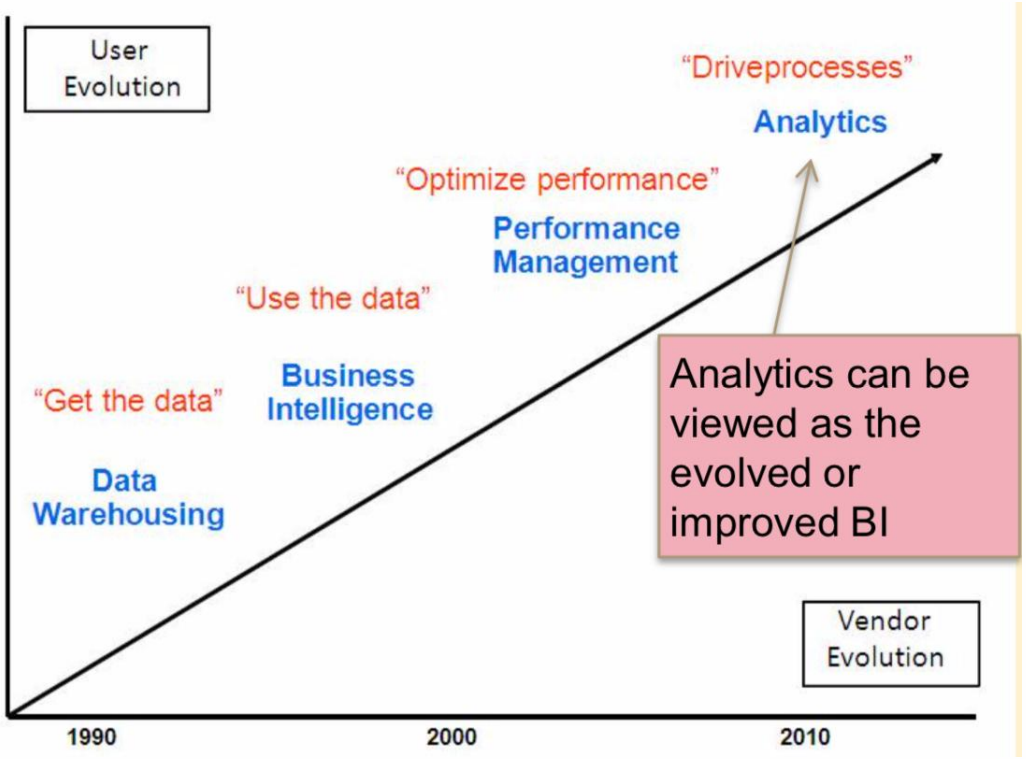
Business Intelligence vs Advanced Analytics



Source: https://www.techtarget.com

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Business Intelligence vs Advanced Analytics (cont'd)



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BI General Process - Data Presentation

- The last mile of BI is the presentation of data or analysis to human users
- Data presentation is the method by which people summarize, organize and communicate information using a variety of tools, including tables, diagrams/charts, and other visualization techniques
- Multiple ways to present results
  - Regular/periodical static reports
  - Interactive and exportable reports
  - Live and real time dashboard
  - Free form ad hoc analysis
  - Edited PowerPoint
- Presentation commonly utilizes data visualization techniques to assist interpreting and presenting data in a visual way.

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BI General Process - Data Presentation (Report and Dashboard)

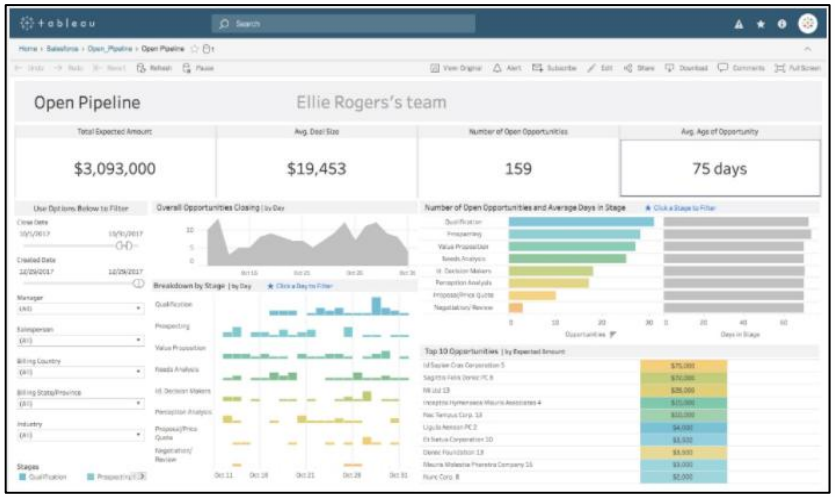
- **Report**
  - A report is the presentation of detailed data arranged in defined layouts and formats.
  - Based on simple and direct queries – usually involves simple analysis and transformation of data (sorting, calculating, filtering, grouping, formatting, etc.)
- Traditional reports contain detailed data in a tabular format and typically display numbers and texts only.
  - It is geared towards people who need data rather than a direct understanding or interpretation of data.
  - Its purpose is mainly for printing (with styling) or exporting (raw data).
- Modern reports can be interactive and visual but the focus is still on detailed data. Sometimes the distinction is a bit blurred with dashboards in some practical cases.

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BI General Process - Data Presentation (Report and Dashboard)

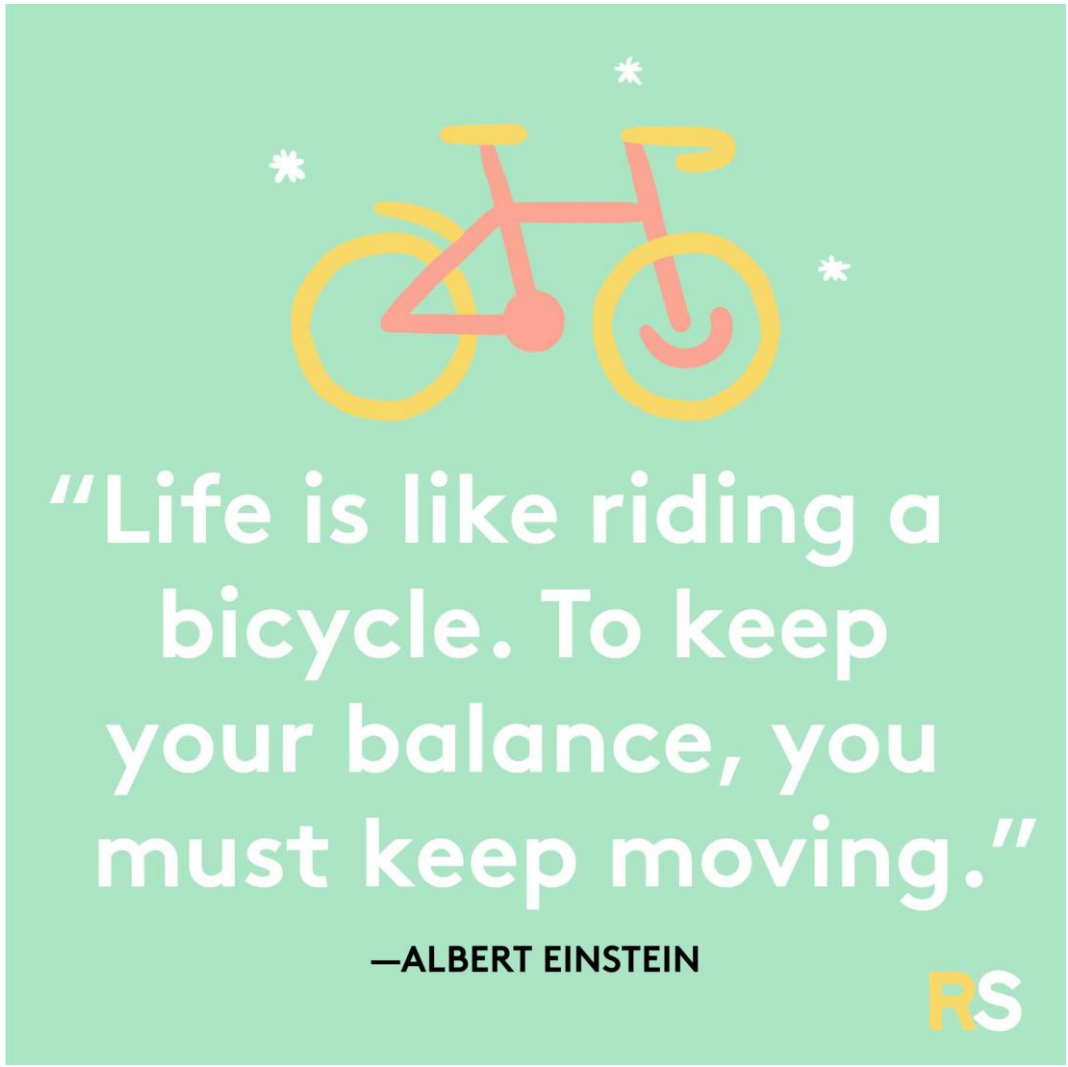
- **Dashboard**
  - A dashboard is a visual-oriented display of the most important data and information needed to achieve defined goals and objectives; consolidated and arranged on a single screen so the information can be viewed at a glance.

    (Stephen Few, Dashboard Confusion, http://www.perceptualedge.com/articles/ie/dashboard_confusion.pdf





Sample dashboard


"Life is like riding a bicycle. To keep your balance, you must keep moving."
—ALBERT EINSTEIN
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